Verification of Underworld3 spherical-shell Stokes flow with the Kramer benchmark

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Abstract. We reproduce the spherical-shell Stokes benchmarks of Kramer et al. (2021) using Underworld3 (UW3). The benchmarks provide analytical velocity and pressure solutions for incompressible, isoviscous mantle flow in a three-dimensional spherical shell and include both smooth volumetric density forcing and delta-function forcing on an internal spherical interface. We test free-slip and zero-slip boundary conditions on tetrahedral spherical-shell meshes with a $P_2 \times P_1$ Taylor–Hood discretisation. Numerical velocity, pressure, boundary velocity, and radial normal-stress errors are compared with the analytical fields using relative L_2-norm metrics. Smooth pressure errors converge at approximately second order, while smooth velocity errors converge closer to second order than to the ideal third-order P_2 rate. Delta-function cases show the expected reduced rates, approximately $O(h^{1.5})$ for velocity and $O(h^{0.5})$ for pressure. Boundary velocity and radial-stress diagnostics decrease systematically but show stronger pre-asymptotic and regularity effects. The results verify the main spherical Kramer benchmark trends in UW3 and identify smooth-case velocity convergence and boundary stress recovery as the principal targets for further curved-shell verification.

1 Introduction

Analytical Stokes benchmarks are central to verifying geodynamic finite-element codes because they test velocity–pressure coupling, incompressibility, boundary-condition enforcement, pressure normalisation, and stress recovery without relying on code-to-code comparison. These issues are especially important in spherical geometry, where curved boundaries and spherical harmonic forcing interact with mesh geometry and boundary quadrature.

The benchmark solutions of Kramer et al. (2021) extend geodynamic Stokes verification to cylindrical and spherical shells. In the spherical case they provide exact solutions for smooth density forcing and for singular forcing on an internal interface. This distinction is useful: the smooth cases test the asymptotic behaviour expected for polynomial finite elements, while the delta-function cases deliberately reduce solution regularity and therefore test whether the numerical method reproduces the weaker rates predicted by the benchmark.

Here we present a concise UW3 verification article for the spherical Kramer benchmark. The calculations use existing benchmark outputs only; no figures or tables are regenerated. The manuscript follows the structure of the annulus Copernicus-style verification article and condenses the corresponding Kramer spherical report into the governing equations, benchmark definition, numerical setup, error metrics, results, discussion, and conclusions.

Table 1. Kramer spherical benchmark cases used in the UW3 verification runs.

Case	Forcing	Boundary condition	(l,m,k) values
1	Delta-function	Free-slip	$(2,1,3)$, $(2,2,3)$, $(4,2,5)$, $(4,4,5)$, $(8,4,9)$, $(8,8,9)$
2	Smooth	Free-slip	same as Case 1
3	Delta-function	Zero-slip	same as Case 1
4	Smooth	Zero-slip	same as Case 1

2 Governing equations and benchmark solution

The calculations solve the steady incompressible Stokes equations

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{1b}$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration. The Cauchy stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta\boldsymbol{\varepsilon}(\mathbf{u}), \qquad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right), \tag{2}$$

with constant viscosity η.

The domain is a spherical shell

$$\Omega = \left\{\mathbf{x} \in \mathbb{R}^3 : R_- \le r \le R_+\right\}, \qquad R_- = 1.22, \qquad R_+ = 2.22, \tag{3}$$

where $r = \|\mathbf{x}\|$. Delta-function cases use the internal spherical interface

$$R_{\text{int}} = 2.0 = \frac{R_- + R_+}{2}. \tag{4}$$

Following Kramer et al. (2021), the forcing is expressed with spherical harmonics Y_l^m. Smooth cases prescribe

$$\rho(r,\theta,\phi) = \left(\frac{r}{R_+}\right)^k Y_l^m(\theta,\phi), \qquad k = l+1, \tag{5}$$

whereas delta-function cases prescribe an interface-localised forcing proportional to

$$\delta(r - R_{\text{int}}) Y_l^m(\theta,\phi). \tag{6}$$

The corresponding analytical velocity and pressure fields are the spherical shell solutions derived by Kramer et al. (2021). In this verification exercise they are used as exact reference fields for volume and boundary error evaluation.

Figure 1 shows the analytical density fields used for the smooth and delta-function forcing families. Increasing l and m increases the angular complexity of the solution, while increasing k in the smooth cases concentrates density variation toward the outer boundary. The delta-function cases are less regular because the forcing is localised on R_{int}.

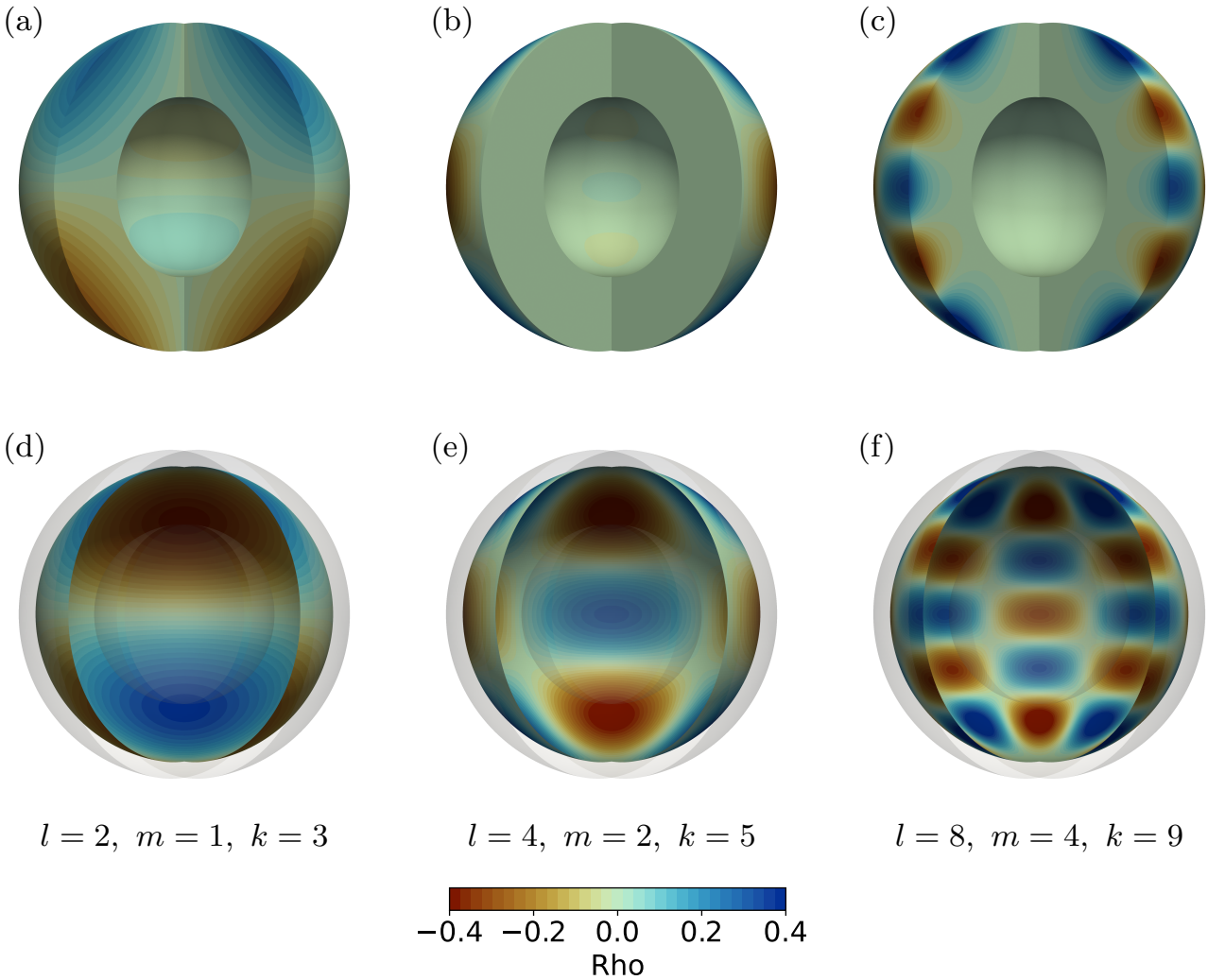

Figure 1. Analytical density fields. The upper row shows smooth forcing and the lower row shows delta-function forcing on $R_{\mathrm{int}} = 2.0$. Columns correspond to $(l, m, k) = (2, 1, 3)$, $(4, 2, 5)$, and $(8, 4, 9)$.

3 Numerical setup

The UW3 calculations use quadratic velocity and linear pressure finite elements on tetrahedral spherical-shell meshes, i.e. a $P_2 \times P_1$ Taylor–Hood pair. The mesh sizes are

$h = 1/8,\ 1/16,\ 1/32,\ 1/64.$

Free-slip boundary conditions are imposed weakly, while zero-slip cases use essential velocity constraints. Solver tolerances are chosen so that the reported errors are dominated by discretisation rather than by linear-solver residuals. As in any incom-

pressible Stokes solve, the pressure is determined only up to an additive constant; the numerical pressure is normalised before comparison with the analytical reference pressure.

4 Error metrics

Volume errors are measured with relative L_2 norms. For a scalar or vector quantity q, with analytical reference q^*, the error is

$$E_{L_2}^{\mathrm{rel}}(q) = \left(\frac{\int_\Omega |q - q^*|^2 \, \mathrm{d}\Omega}{\int_\Omega |q^*|^2 \, \mathrm{d}\Omega} \right)^{1/2}. \tag{7}$$

Boundary diagnostics use the same relative norm on the inner or outer spherical surface. We write these surfaces as Γ_{inner} and Γ_{outer}, so the table and figure labels use $E_{L_2,\Gamma_{\mathrm{inner}}}^{\mathrm{rel}}(\cdot)$ and $E_{L_2,\Gamma_{\mathrm{outer}}}^{\mathrm{rel}}(\cdot)$:

$$E_{L_2,\Gamma}^{\mathrm{rel}}(q) = \left(\frac{\int_\Gamma |q - q^*|^2 \, \mathrm{d}\Gamma}{\int_\Gamma |q^*|^2 \, \mathrm{d}\Gamma} \right)^{1/2}. \tag{8}$$

Observed convergence rates are computed from successive mesh refinements as

$$\alpha = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}, \tag{9}$$

which reduces to $\alpha = \log_2(E_h/E_{h/2})$ for the refinements used here.

5 Results

5.1 Volume velocity and pressure errors

For smooth Stokes solutions, a $P_2 \times P_1$ Taylor–Hood method with accurately represented curved geometry is expected to approach third-order velocity convergence and second-order pressure convergence in the L_2 norm (Girault and Raviart, 1986; Boffi et al., 2013). The UW3 smooth spherical cases reproduce the second-order pressure trend but do not reach the ideal third-order velocity rate over the tested resolutions. Case 2, with smooth forcing and free-slip boundaries, gives velocity rates mostly between 1.9 and 2.1. Case 4, with smooth forcing and zero-slip boundaries, is similarly close to second order, apart from higher coarse-grid pre-asymptotic rates for some higher spherical harmonics.

The delta-function cases reproduce the reduced regularity trend reported by Kramer et al. (2021). In Cases 1 and 3, the finest-grid velocity rates are close to $O(h^{1.5})$, while pressure rates are close to $O(h^{0.5})$. This behaviour is expected because the interface-localised forcing limits the regularity of both velocity gradients and pressure. It is therefore a benchmark feature rather than a solver failure.

The numerical values and pairwise rates for Fig. 2 are included as existing benchmark tables after the main text.

5.2 Boundary velocity and radial normal stress

Boundary metrics are evaluated on the inner and outer spherical surfaces for Case 1, the free-slip delta-function benchmark. Boundary velocity errors fall rapidly on the coarse refinements, with outer-boundary rates near third to fourth order. On the

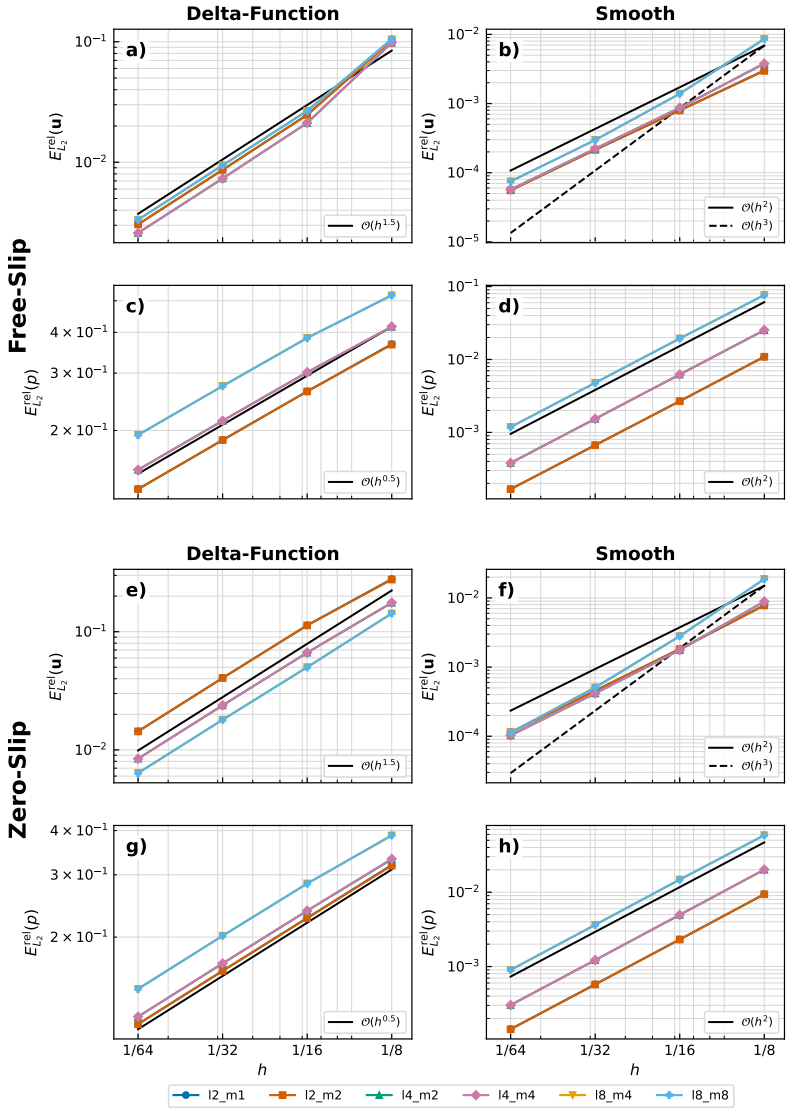

Figure 2. Velocity and pressure convergence for the four spherical Kramer cases. Reference slopes distinguish smooth-solution expectations from the reduced rates associated with delta-function forcing.

finest refinement the inner-boundary velocity rates reduce to approximately 2.3–2.6, while the outer boundary remains higher for the lower spherical harmonics.

The radial normal stress σ_{rr} is more sensitive than velocity because it combines pressure and velocity-gradient errors and is then evaluated on a curved boundary. The stress errors decrease systematically but show more variable rates than the volume norms. Coarse-grid rates are partly pre-asymptotic, whereas finest-grid rates are closer to first-to-second order. This is

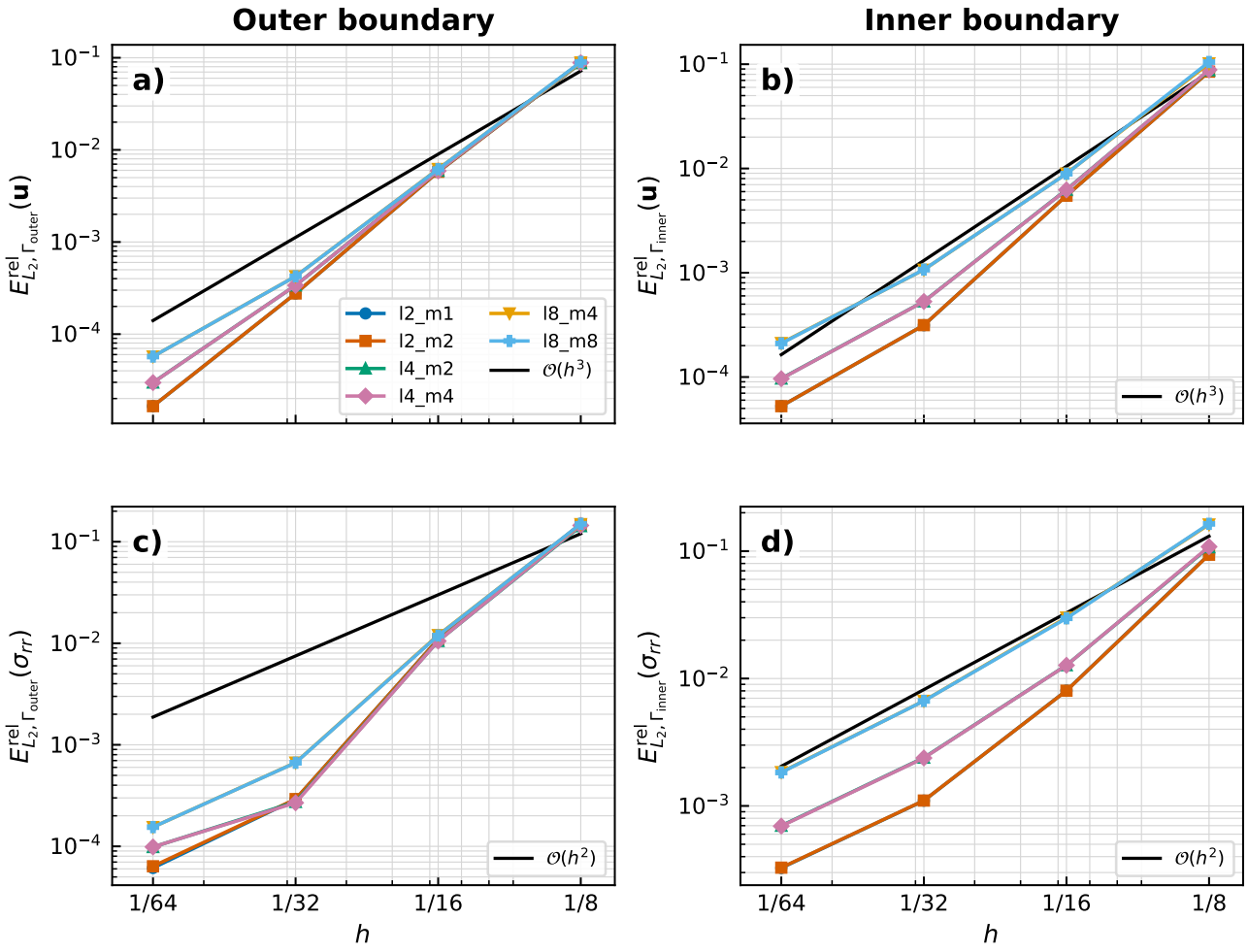

Figure 3. Boundary velocity and radial normal-stress convergence on the inner and outer spherical boundaries for Case 1.

consistent with the singular forcing, continuous pressure approximation, derivative recovery, boundary quadrature, and weak free-slip enforcement all contributing to the boundary stress error.

The corresponding boundary velocity and radial normal-stress values are included as existing benchmark tables after the main text.

6 Discussion

The UW3 calculations reproduce the primary Kramer spherical benchmark distinction between smooth and delta-function forcing. Smooth pressure converges close to the expected $O(h^2)$ rate, and delta-function forcing reduces velocity and pres-

sure convergence to the expected approximate $O(h^{1.5})$ and $O(h^{0.5})$ trends. These results indicate that the analytical forcing, pressure normalisation, and mixed Stokes solve are implemented consistently.

The main discrepancy from the ideal smooth-solution expectation is velocity: the smooth cases converge closer to second order than to the third-order P_2 L_2 rate obtained by Kramer et al. (2021) with curved/isoparametric Fluidity calculations. The pressure result narrows the likely explanation. The Stokes solve and smooth forcing are behaving consistently, but geometry representation, boundary-condition enforcement, quadrature, mesh quality, or the interaction among these terms appears to limit the velocity error before the polynomial interpolation rate is reached.

The boundary results are intentionally more demanding than volume norms. Radial normal stress contains pressure directly and velocity derivatives through $2\eta\boldsymbol{\varepsilon}(\mathbf{u})$, so it inherits pressure regularity limits and derivative-recovery errors. Differences between inner and outer boundary errors are therefore expected because the surfaces have different radii, analytical amplitudes, quadrature weights, and sensitivity to mesh geometry. Higher (l, m) values further increase angular gradients and delay asymptotic behaviour.

7 Conclusions

We have produced a Copernicus-style verification article for the UW3 implementation of the Kramer spherical Stokes benchmark. The principal conclusions are:

1. UW3 reproduces the expected contrast between smooth volumetric forcing and delta-function interface forcing in a spherical shell.
2. Smooth pressure errors converge at approximately second order for the $P_2 \times P_1$ Taylor–Hood discretisation.
3. Smooth velocity errors converge robustly but remain closer to second order than to the ideal third-order P_2 L_2 rate over the tested resolutions.
4. Delta-function cases show the expected reduced convergence rates, approximately $O(h^{1.5})$ for velocity and $O(h^{0.5})$ for pressure.
5. Boundary velocity and radial normal-stress errors decrease systematically, with stress recovery showing the strongest sensitivity to singular forcing and curved-boundary post-processing.

Overall, the benchmark supports UW3 spherical-shell Stokes verification while identifying smooth-case velocity convergence and boundary stress recovery as the most useful next diagnostics for curved geometry and boundary-condition development.

Benchmark tables

Tables 2–9 list the volume velocity and pressure errors and pairwise rates for the spherical Kramer cases. Tables 10–13 list the Case 1 boundary velocity and radial normal-stress errors.

Table 2. Case 1: free-slip, delta-function forcing (1 of 2).

h	$l=2, m=1, k=3$				$l=2, m=2, k=3$				$l=4, m=2, k=5$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	1.034e-01	–	3.683e-01	–	1.024e-01	–	3.669e-01	–	9.793e-02	–	4.161e-01	–
$1/16$	2.473e-02	2.06	2.636e-01	0.48	2.470e-02	2.05	2.635e-01	0.48	2.117e-02	2.21	3.017e-01	0.46
$1/32$	8.648e-03	1.52	1.869e-01	0.50	8.644e-03	1.51	1.869e-01	0.50	7.336e-03	1.53	2.142e-01	0.49
$1/64$	3.053e-03	1.50	1.321e-01	0.50	3.052e-03	1.50	1.321e-01	0.50	2.591e-03	1.50	1.515e-01	0.50

Table 3. Case 1: free-slip, delta-function forcing (2 of 2).

h	$l=4, m=4, k=5$				$l=8, m=4, k=9$				$l=8, m=8, k=9$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	9.858e-02	–	4.169e-01	–	1.045e-01	–	5.192e-01	–	1.055e-01	–	5.206e-01	–
$1/16$	2.119e-02	2.22	3.019e-01	0.47	2.674e-02	1.97	3.846e-01	0.43	2.666e-02	1.98	3.843e-01	0.44
$1/32$	7.335e-03	1.53	2.142e-01	0.50	9.438e-03	1.50	2.738e-01	0.49	9.420e-03	1.50	2.737e-01	0.49
$1/64$	2.589e-03	1.50	1.514e-01	0.50	3.336e-03	1.50	1.938e-01	0.50	3.333e-03	1.50	1.937e-01	0.50

Table 4. Case 2: free-slip, smooth forcing (1 of 2).

h	$l=2, m=1, k=3$				$l=2, m=2, k=3$				$l=4, m=2, k=5$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	3.011e-03	–	1.090e-02	–	2.964e-03	–	1.088e-02	–	3.803e-03	–	2.519e-02	–
$1/16$	7.961e-04	1.92	2.681e-03	2.02	7.957e-04	1.90	2.680e-03	2.02	8.684e-04	2.13	6.219e-03	2.02
$1/32$	2.145e-04	1.89	6.681e-04	2.00	2.145e-04	1.89	6.678e-04	2.00	2.221e-04	1.97	1.536e-03	2.02
$1/64$	5.570e-05	1.95	1.669e-04	2.00	5.568e-05	1.95	1.667e-04	2.00	5.706e-05	1.96	3.813e-04	2.01

Table 5. Case 2: free-slip, smooth forcing (2 of 2).

h	$l=4, m=4, k=5$				$l=8, m=4, k=9$				$l=8, m=8, k=9$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	3.778e-03	–	2.534e-02	–	8.580e-03	–	7.633e-02	–	8.597e-03	–	7.634e-02	–
$1/16$	8.697e-04	2.12	6.221e-03	2.03	1.384e-03	2.63	1.937e-02	1.98	1.387e-03	2.63	1.940e-02	1.98
$1/32$	2.221e-04	1.97	1.534e-03	2.02	2.969e-04	2.22	4.787e-03	2.02	2.968e-04	2.22	4.776e-03	2.02
$1/64$	5.702e-05	1.96	3.803e-04	2.01	7.478e-05	1.99	1.184e-03	2.02	7.484e-05	1.99	1.183e-03	2.01

Table 6. Case 3: zero-slip, delta-function forcing (1 of 2).

h	$l=2, m=1, k=3$				$l=2, m=2, k=3$				$l=4, m=2, k=5$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	2.791e-01	–	3.198e-01	–	2.762e-01	–	3.185e-01	–	1.757e-01	–	3.320e-01	–
$1/16$	1.133e-01	1.30	2.265e-01	0.50	1.131e-01	1.29	2.264e-01	0.49	6.647e-02	1.40	2.375e-01	0.48
$1/32$	4.060e-02	1.48	1.606e-01	0.50	4.058e-02	1.48	1.605e-01	0.50	2.382e-02	1.48	1.685e-01	0.49
$1/64$	1.434e-02	1.50	1.135e-01	0.50	1.434e-02	1.50	1.135e-01	0.50	8.421e-03	1.50	1.192e-01	0.50

Table 7. Case 3: zero-slip, delta-function forcing (2 of 2).

h	$l=4, m=4, k=5$				$l=8, m=4, k=9$				$l=8, m=8, k=9$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	1.765e-01	–	3.326e-01	–	1.422e-01	–	3.871e-01	–	1.432e-01	–	3.882e-01	–
$1/16$	6.656e-02	1.41	2.375e-01	0.49	5.011e-02	1.51	2.834e-01	0.45	4.997e-02	1.52	2.832e-01	0.46
$1/32$	2.382e-02	1.48	1.685e-01	0.50	1.800e-02	1.48	2.018e-01	0.49	1.796e-02	1.48	2.017e-01	0.49
$1/64$	8.415e-03	1.50	1.192e-01	0.50	6.366e-03	1.50	1.428e-01	0.50	6.359e-03	1.50	1.427e-01	0.50

Table 8. Case 4: zero-slip, smooth forcing (1 of 2).

h	$l=2, m=1, k=3$				$l=2, m=2, k=3$				$l=4, m=2, k=5$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	7.826e-03	–	9.412e-03	–	7.788e-03	–	9.399e-03	–	8.891e-03	–	2.009e-02	–
$1/16$	1.826e-03	2.10	2.313e-03	2.02	1.820e-03	2.10	2.311e-03	2.02	1.764e-03	2.33	4.960e-03	2.02
$1/32$	4.559e-04	2.00	5.754e-04	2.01	4.553e-04	2.00	5.744e-04	2.01	4.164e-04	2.08	1.223e-03	2.02
$1/64$	1.141e-04	2.00	1.435e-04	2.00	1.140e-04	2.00	1.432e-04	2.00	1.029e-04	2.02	3.032e-04	2.01

Table 9. Case 4: zero-slip, smooth forcing (2 of 2).

h	$l=4, m=4, k=5$				$l=8, m=4, k=9$				$l=8, m=8, k=9$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	8.873e-03	–	2.021e-02	–	1.873e-02	–	5.851e-02	–	1.875e-02	–	5.853e-02	–
$1/16$	1.763e-03	2.33	4.960e-03	2.03	2.801e-03	2.74	1.477e-02	1.99	2.798e-03	2.74	1.480e-02	1.98
$1/32$	4.164e-04	2.08	1.222e-03	2.02	5.075e-04	2.46	3.645e-03	2.02	5.073e-04	2.46	3.635e-03	2.03
$1/64$	1.028e-04	2.02	3.026e-04	2.01	1.130e-04	2.17	9.003e-04	2.02	1.129e-04	2.17	8.992e-04	2.02

Table 10. Case 1 boundary velocity convergence (1 of 2).

h	$l=2, m=1, k=3$				$l=2, m=2, k=3$				$l=4, m=2, k=5$			
	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\mathbf{u})$	rate
$1/8$	8.877e-02	–	8.511e-02	–	8.782e-02	–	8.478e-02	–	8.760e-02	–	8.756e-02	–
$1/16$	5.788e-03	3.94	5.496e-03	3.95	5.786e-03	3.92	5.487e-03	3.95	5.942e-03	3.88	6.265e-03	3.80
$1/32$	2.762e-04	4.39	3.142e-04	4.13	2.764e-04	4.39	3.145e-04	4.12	3.383e-04	4.13	5.312e-04	3.56
$1/64$	1.653e-05	4.06	5.257e-05	2.58	1.654e-05	4.06	5.257e-05	2.58	2.984e-05	3.50	9.674e-05	2.46

Table 11. Case 1 boundary velocity convergence (2 of 2).

h	$l=4, m=4, k=5$				$l=8, m=4, k=9$				$l=8, m=8, k=9$			
	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\mathbf{u})$	rate
$1/8$	8.839e-02	–	8.836e-02	–	8.885e-02	–	1.018e-01	–	8.986e-02	–	1.049e-01	–
$1/16$	5.913e-03	3.90	6.234e-03	3.83	6.205e-03	3.84	8.936e-03	3.51	6.170e-03	3.86	8.911e-03	3.56
$1/32$	3.382e-04	4.13	5.296e-04	3.56	4.244e-04	3.87	1.069e-03	3.06	4.239e-04	3.86	1.072e-03	3.05
$1/64$	2.976e-05	3.51	9.631e-05	2.46	5.715e-05	2.89	2.117e-04	2.34	5.713e-05	2.89	2.100e-04	2.35

Table 12. Case 1 boundary radial-stress convergence (1 of 2).

h	$l=2, m=1, k=3$				$l=2, m=2, k=3$				$l=4, m=2, k=5$			
	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	rate
$1/8$	1.500e-01	–	9.413e-02	–	1.458e-01	–	9.370e-02	–	1.424e-01	–	1.073e-01	–
$1/16$	1.096e-02	3.77	8.041e-03	3.55	1.095e-02	3.74	8.024e-03	3.55	1.054e-02	3.76	1.272e-02	3.08
$1/32$	2.905e-04	5.24	1.100e-03	2.87	2.920e-04	5.23	1.102e-03	2.86	2.737e-04	5.27	2.392e-03	2.41
$1/64$	6.125e-05	2.25	3.258e-04	1.76	6.364e-05	2.20	3.276e-04	1.75	9.848e-05	1.47	6.990e-04	1.77

Table 13. Case 1 boundary radial-stress convergence (2 of 2).

h	$l=4, m=4, k=5$				$l=8, m=4, k=9$				$l=8, m=8, k=9$			
	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{outer}}}(\sigma_{rr})$	rate	$E^{\mathrm{rel}}_{L_2,\Gamma_{\mathrm{inner}}}(\sigma_{rr})$	rate
$1/8$	1.447e-01	–	1.083e-01	–	1.489e-01	–	1.618e-01	–	1.499e-01	–	1.637e-01	–
$1/16$	1.052e-02	3.78	1.270e-02	3.09	1.203e-02	3.63	3.009e-02	2.43	1.192e-02	3.65	2.971e-02	2.46
$1/32$	2.683e-04	5.29	2.379e-03	2.42	6.680e-04	4.17	6.678e-03	2.17	6.658e-04	4.16	6.670e-03	2.16
$1/64$	9.866e-05	1.44	6.942e-04	1.78	1.547e-04	2.11	1.841e-03	1.86	1.547e-04	2.11	1.831e-03	1.87

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